BCSE418L
Explainable Artificial Intelligence

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Module 1
Introduction to Explainable Artificial Intelligence

Fundamentals of XAI - Categorization of XAI - Taxonomy of XAI methods for Machine Learning - Machine Learning Interpretability - Causal Model Induction - Causality learning - XAI techniques and limitations

Fundamentals of XAI

XAI- **EXplainable Artificial Intelligence**

What is XAI?

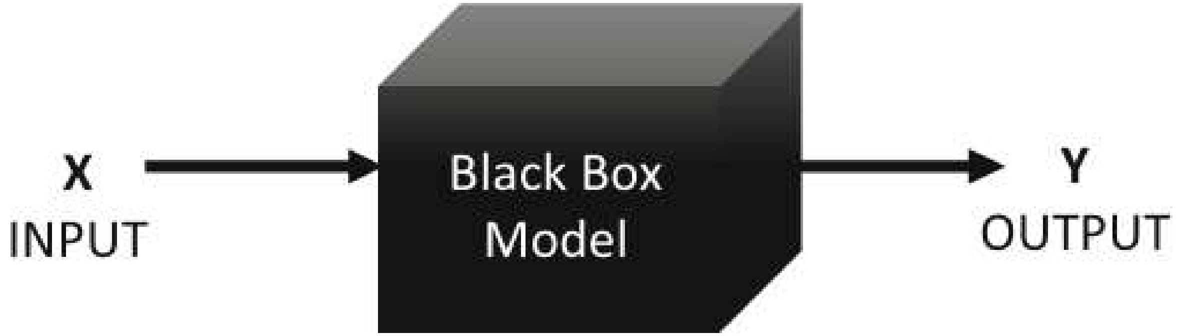
Explainable AI , often overlapping with **interpretable AI** or **Explainable Machine Learning** (**XML**), is a field of research that explores **methods that provide humans with the ability of intellectual oversight over AI algorithms.**

Explainable AI refers to a **set of processes and methods that aim to provide a clear and human-understandable explanation for the decisions generated by AI and machine learning models.**

Fundamentals of XAI

- Explainable AI (XAI) seeks provide us insight on the decision-making ability of an AI system.
- It helps us to understand how, when, and why predictions are made.
- Consequently, it can build greater trust and improve the safety of our use of AI models, encouraging their greater adoption in our society.

Black box model

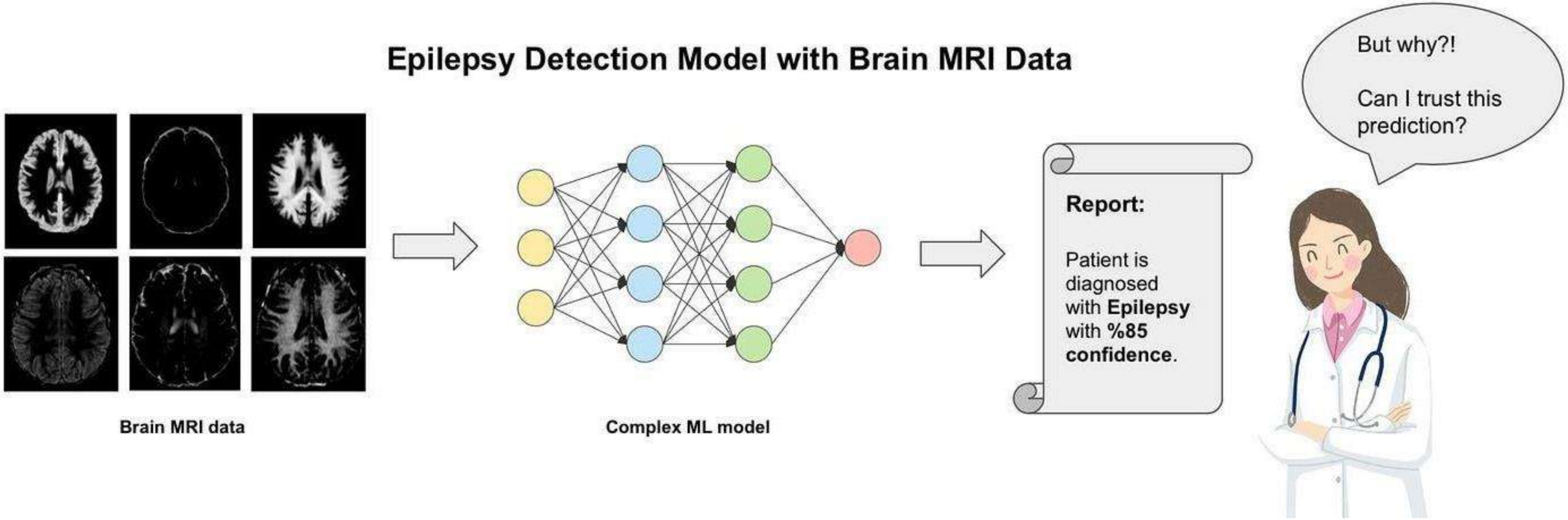


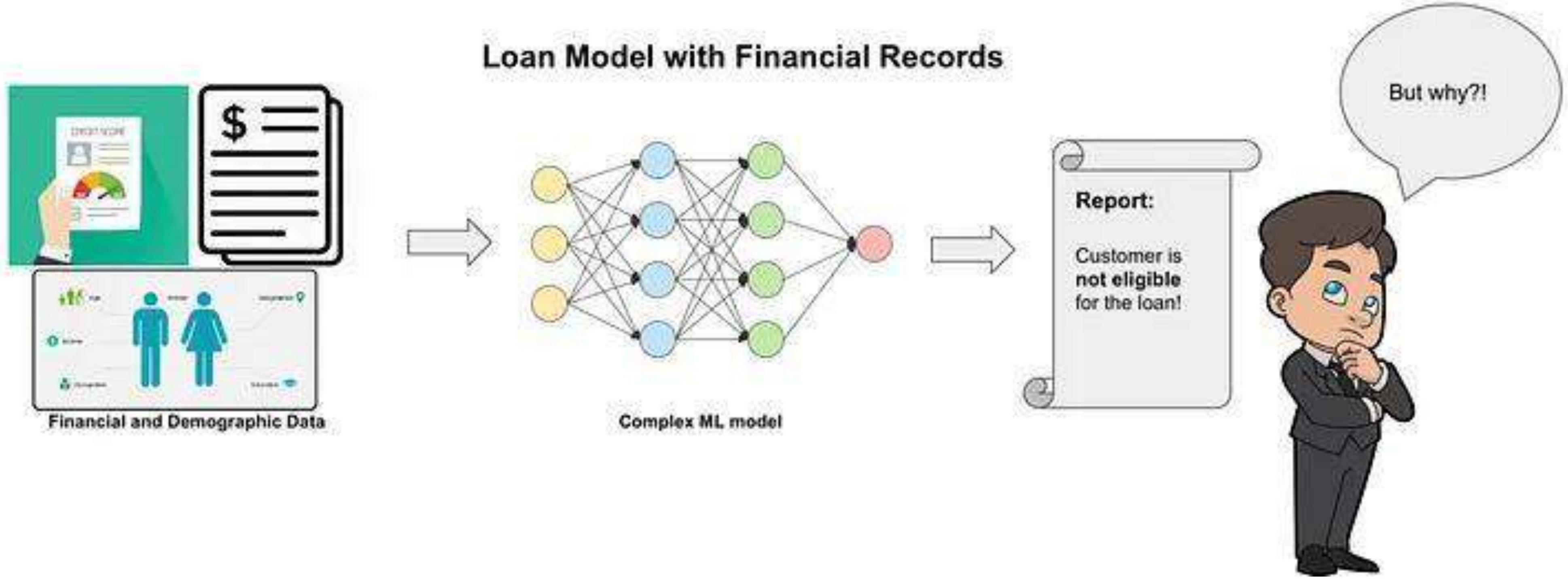
Black-Box Model

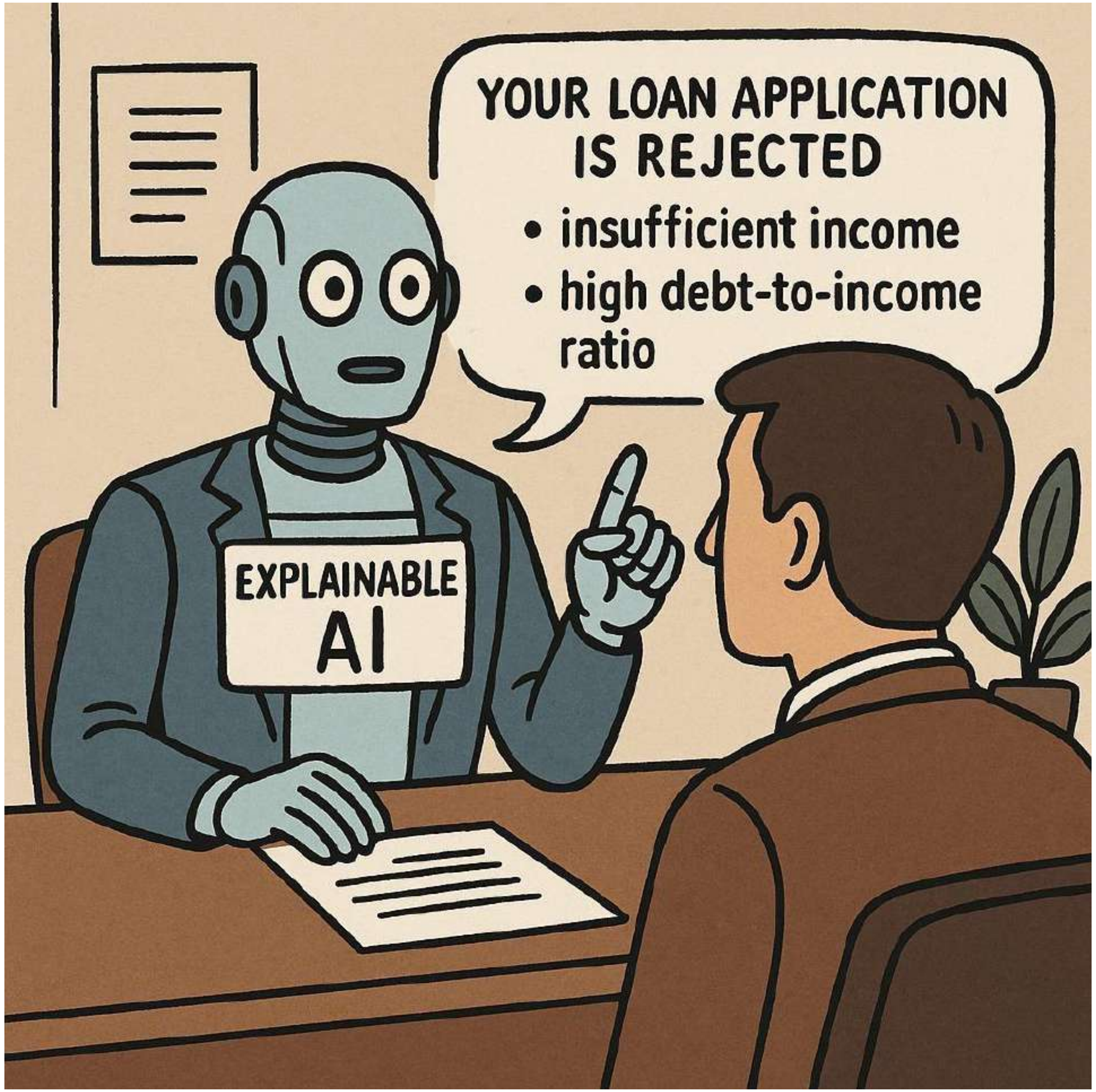
- A black-box model in machine learning refers to model that operates as an opaque system where the internal workings of the model are not easily accessible or interpretable. These models make predictions based on input data, but the decision-making process and reasoning behind the predictions are not transparent to the user .
- This **lack of transparency** makes it difficult for users to understand the model's behavior, detect potential biases or errors, or hold the model accountable for its decisions.
- Example: CNN, RNN,LSTM,SVM etc

Black box model: Example

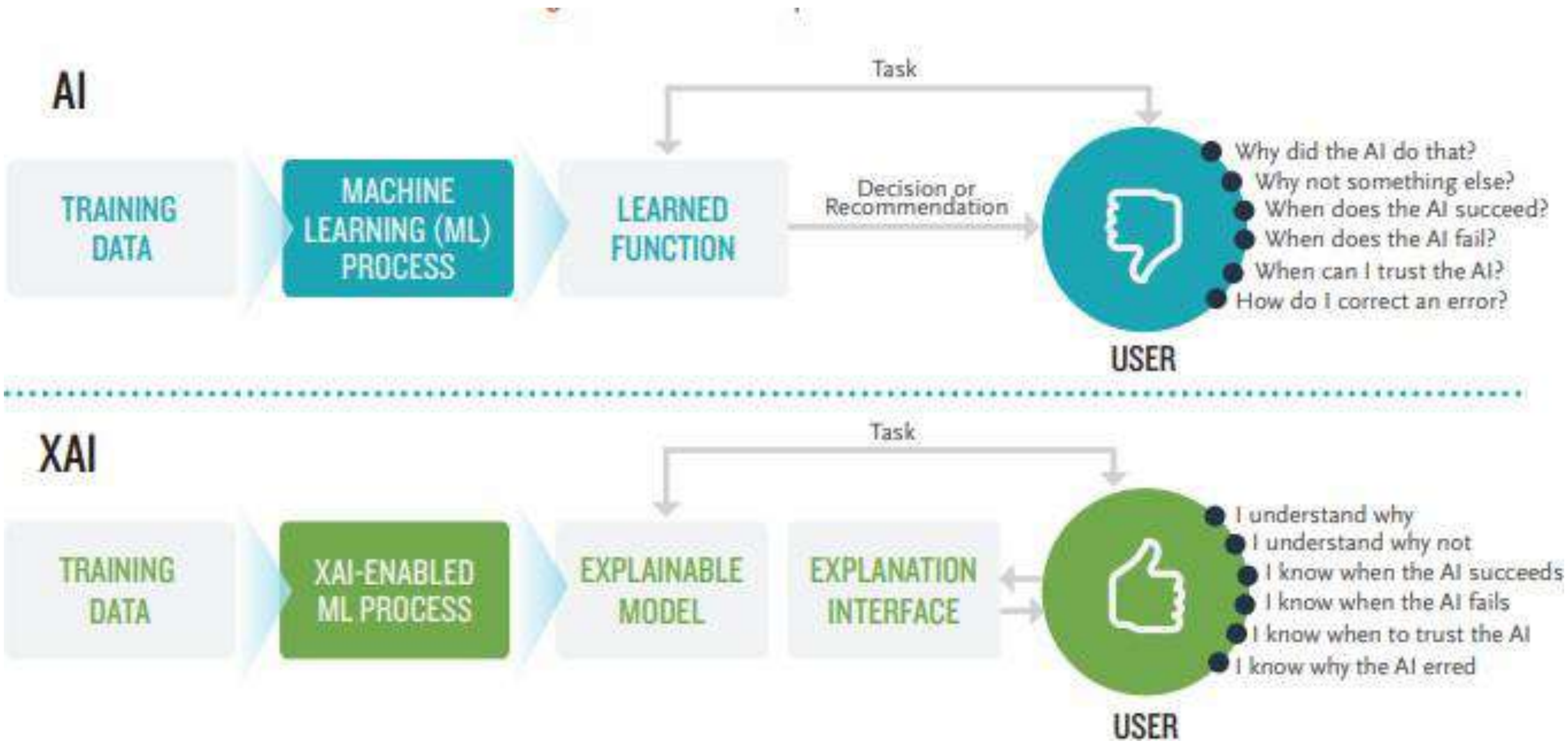
- Consider a black box model that evaluates job candidates' resumes. Users can see the inputs—the resumes they feed into the AI model. And users can see the outputs—the assessments the model returns for those resumes. However, users are often unclear about how the model reaches its conclusions, what factors it considers, and how it weighs those factors.
- Platforms like Netflix, Amazon, and YouTube use Black Box AI to recommend content. The algorithms behind these recommendations are complex and not easily interpretable.
- Virtual assistants like *Siri, Alexa, and Google Assistant* rely on complex AI systems to understand and respond to speech, with internal processes that remain hidden from users.


Epilepsy Detection Model with Brain MRI Data
Brain MRI data
Complex ML model
Report:
Patient is diagnosed with **Epilepsy** with **%85 confidence**.
But why?!
Can I trust this prediction?

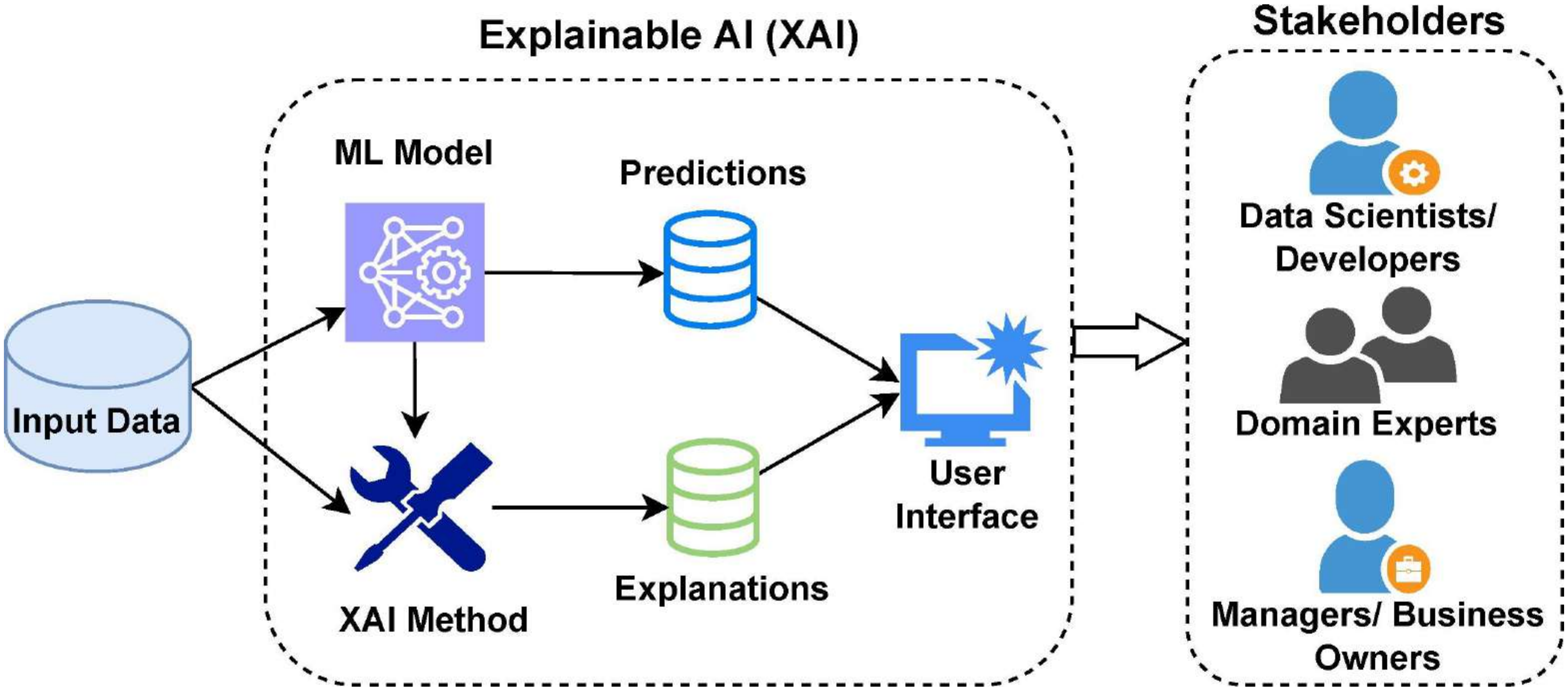

Loan Model with Financial Records
Financial and Demographic Data
Complex ML model
Report:
Customer is **not eligible** for the loan!
But why?!


YOUR LOAN APPLICATION IS REJECTED
• insufficient income
• high debt-to-income ratio
EXPLAINABLE AI

Fundamentals of XAI



Fundamentals of XAI



Explainability

- The process or methods used to **generate** an understandable account of **why** a particular model made a given decision—often via post-hoc tools or intrinsic features.
- The process of elucidating or revealing the decision-making mechanisms of models. The user may see how inputs and outputs are mathematically interlinked.
- It relates to the ability to understand why AI models make their decisions.
- The capacity to make automatic interpretations and describe the inner workings of an AI system in human terms is referred to as explainability.
- An explainable technique summarizes the reasons for an AI model's decision.

Interpretability

- Interpretability is the degree to which a human can understand the cause of a decision.
- A method is interpretable if a user can correctly and efficiently predict the method’s results

Decision Tree

IF Income > ₹50,000
THEN Approve
ELSE IF Credit Score > 700
THEN Approve
ELSE
Reject

Understandability

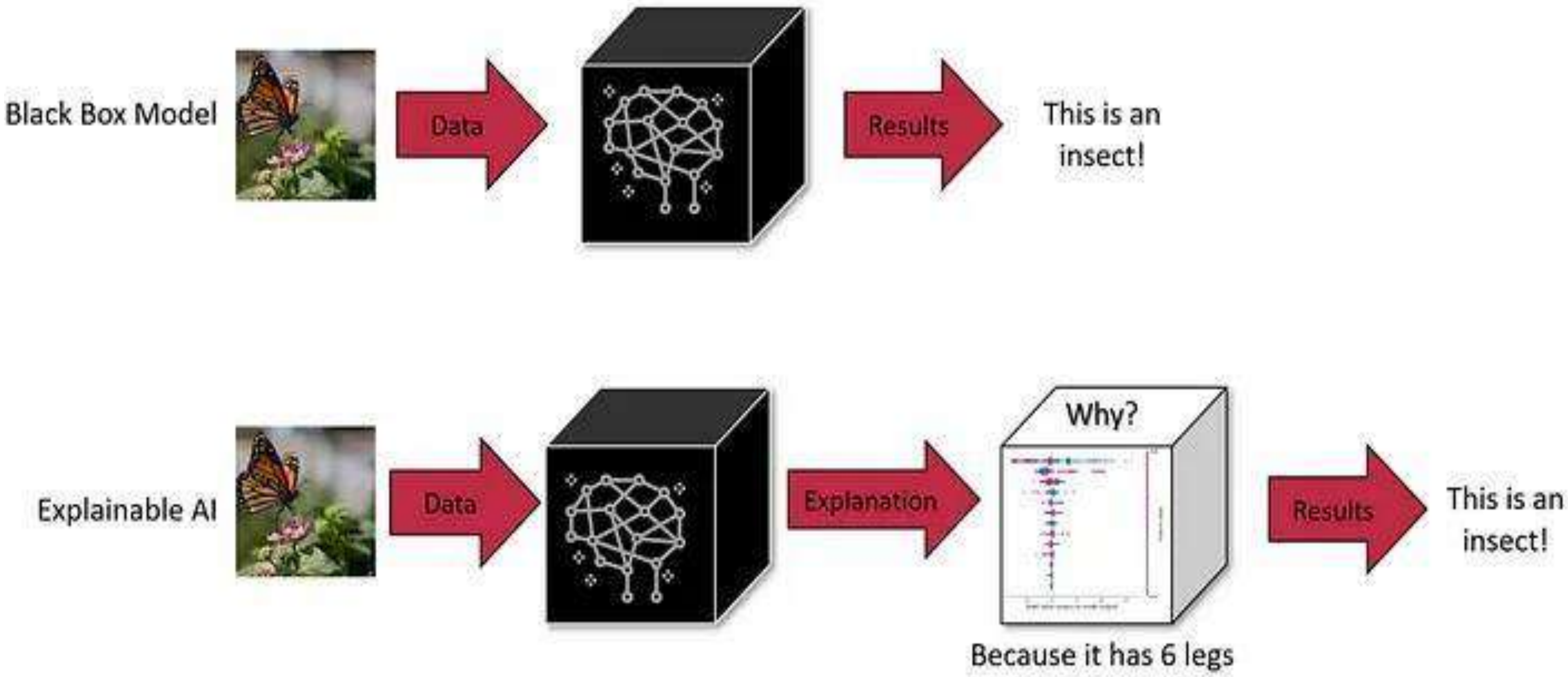
- Understandability is the human comprehension of an explanation.
- How easily a human user (especially a non-expert) can comprehend an explanation—regardless of whether the underlying model is simple or complex.
- Example (Natural-Language Summary):“Your loan was rejected because your income is below the ₹50,000 threshold and your credit score is under 700—even though you have stable employment.”

Transparency

- Transparency is the degree to which a model’s architecture, parameters, and decision-making process are directly accessible and understandable to a human observer.

Example:

- Transparent Model (Decision Tree)
- Opaque Model (Random Forest)



Black Box Model
Data
Results
This is an insect!
Explainable AI
Data
Explanation
Why?
Results
This is an insect!
Because it has 6 legs

White box model

- White box models, also known as transparent or interpretable models, are machine learning models where the internal logic and decision-making process are easily understandable by humans.
- Unlike "black box" models, which are opaque, white box models provide insights into how inputs are transformed into outputs, making it easier to understand, debug, and potentially improve their performance.

Example:

- Decision Trees, Linear Regression, Logistic Regression, Rule-Based Systems

White box model

- A **white-box model** is a machine learning model whose **internal logic is transparent and understandable** to humans.
- You can **see how the model makes decisions**
- You can **trace the influence of input features**
- It supports **explainability, debugging, and trust.**

Why XAI ?

- Learning from multiple problem-solving episodes
- Lack of Transparency
- Not being able to understand System bias
- Not being able to correct the errors in the system
- Improve user trust and system acceptability

Learning from Multiple Problem Solving

- Deep Blue beating Garry Kasparov, but not able to explain its strategy to train other players.
- In 1997, IBM's Deep Blue did something that no machine had done before. In May of that year, it became the first computer system to defeat a reigning world chess champion in a match under standard tournament controls. Big Blue's victory in the six-game marathon against Garry Kasparov marked an inflection point in computing, heralding a future in which supercomputers and artificial intelligence could simulate human thinking.
  - The method chosen for problem solving was shallow and brute force.
  - The system could not learn and represent the problem-solving experience, suitable for providing explanations later.

Lack of Transparency

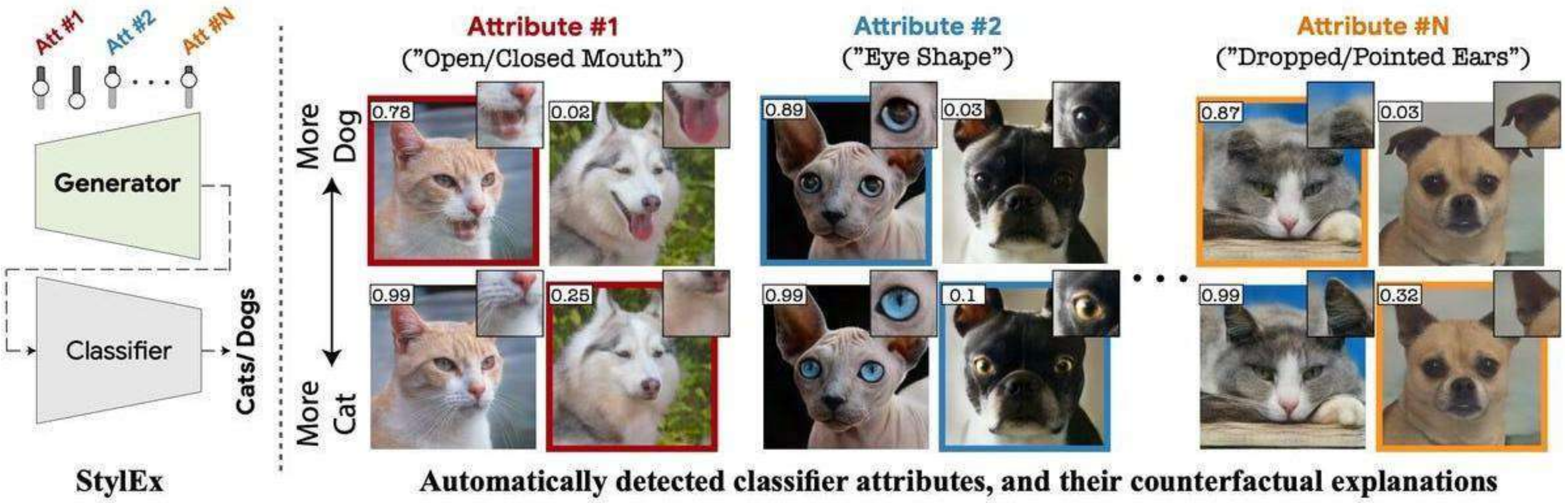
- Using black box models for decision making
  - Cibil Score being reduced
  - Loan being rejected

System Bias

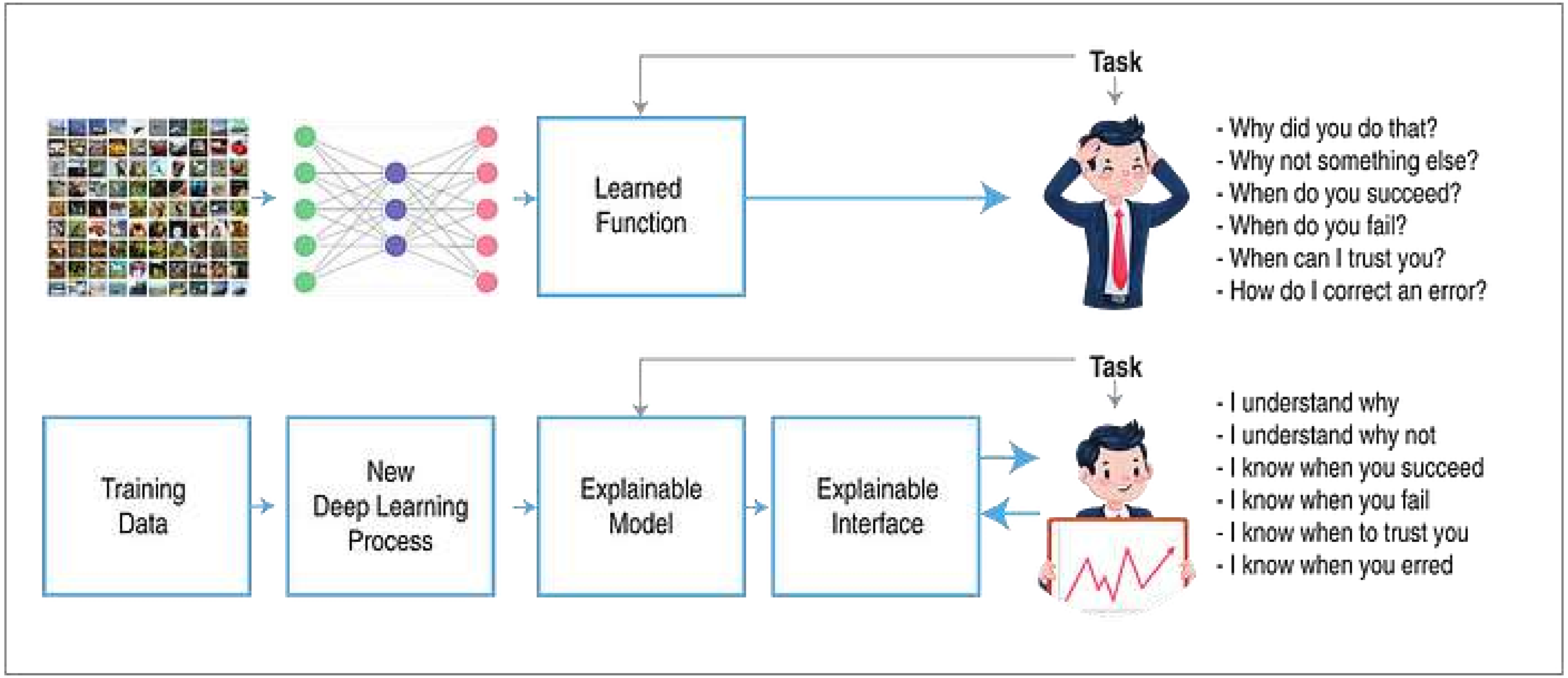
- AI models may be imperfect due to biases
  - **Expert knowledge bias-**Expert knowledge bias, also known as confirmation bias or automation bias, is a type of bias that can occur in machine learning when the assumptions and perspectives of experts, often developers or domain specialists, are unintentionally embedded into the model's design, data, or evaluation, leading to skewed or unfair outcomes.
  - **Data bias-**This occurs when the training data doesn't accurately represent the real-world population the model will encounter, leading to skewed predictions. For example, if a facial recognition system is trained primarily on images of one race, it may perform poorly on other races.
  - **Feature selection bias-**Feature selection bias in machine learning occurs when the process of selecting features for a model introduces a bias, leading to inaccurate or unfair predictions.

Once we visualize the outcome and the logic we may be able to rectify the biases and other errors

Feature Selection



User Acceptance



Features of Explainability

- Fidelity
- Understandability
- Sufficiency
- Low overload
- Efficiency

Fidelity

- Fidelity is the correctness of an ad-hoc technique in generating the true explanations (e.g., correctness of a saliency map) for model predictions.
- In Explainable AI (XAI), fidelity refers to how accurately an explanation reflects the true behavior of the underlying AI model being explained.
- It's a measure of how truthful and reliable the explanation is in representing the model's decision-making process.
- High fidelity means the explanation closely aligns with the model's actual workings, while low fidelity suggests the explanation might be inaccurate or misleading.

Understandability

- Terminology
- User sensitivity
  - User’s Knowledge
  - Goal
  - Preference
  - Concern
- Abstraction
- Summarization
- Perspectives
- Linguistic competence
- feedback

Sufficiency

- Necessity and sufficiency are the building blocks of all successful explanations.
- Sufficiency refers to the extent to which a feature or set of features, when present, guarantees a particular outcome from a machine learning model.

Explanations must include:

- Explanations about system behaviour"
  - How the system solved the problem?
  - How a parameter affected the outcome?
  - What would be the effect due to the change in some of the data?
- Justifications
- Preferences
- Descriptions about the domain
- Terminology definitions

XAI Categories

- The ‘classical’ period of progress in AI — from the 1970s to early 1990s
  - initial phase of interest in methods for explanation generation
  - in symbolic reasoning systems called expert systems
- The recent rapid progress in machine learning (ML), and deep learning has led to a resurgence in interest in explainability.
  - Issues of transparency and accountability have been highlighted as specific areas of concern
- Autonomous Systems where multiple agents collaborate to achieve a task.

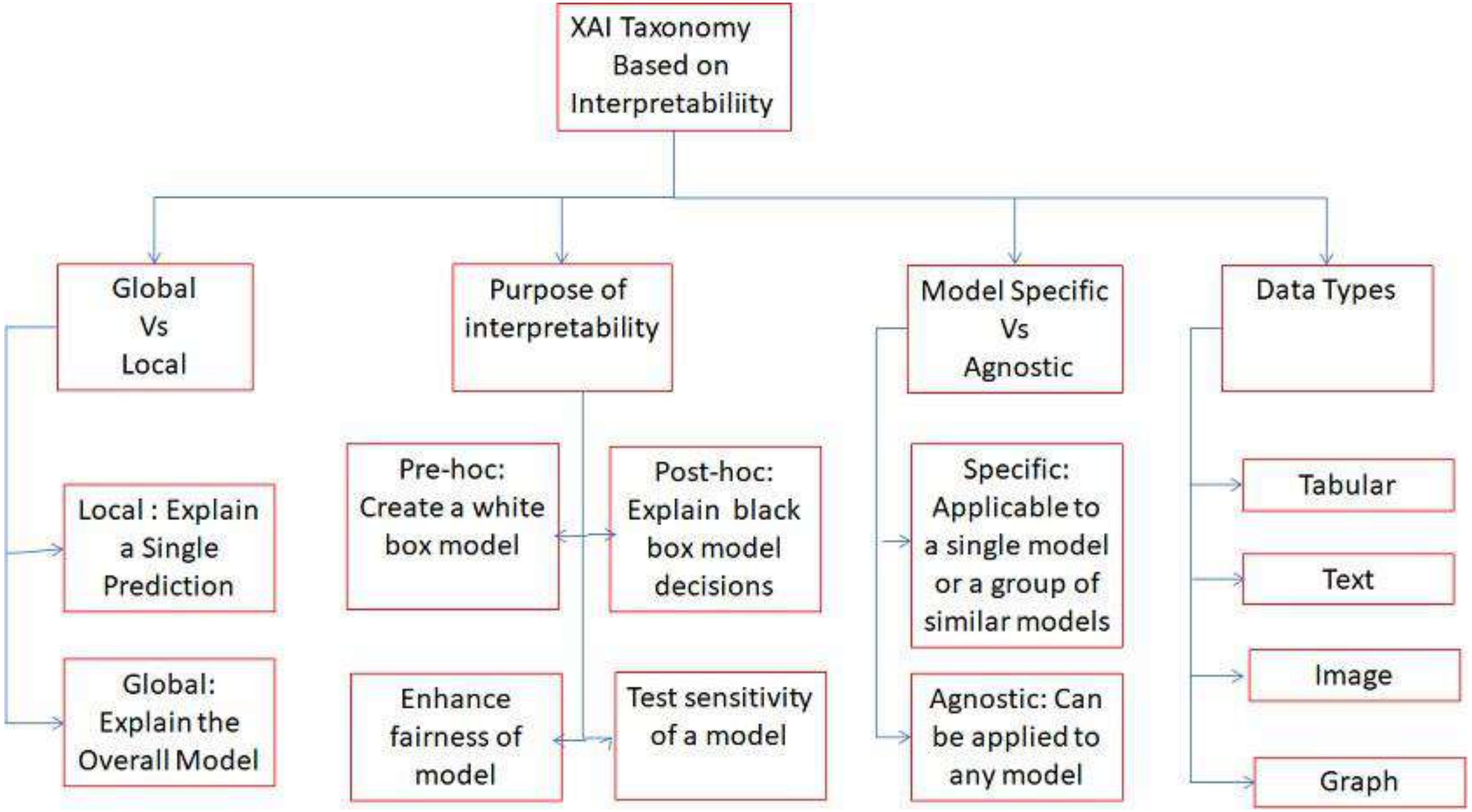
Current XAI Categories

- Goal-Oriented XAI
  - Explanations in Autonomous Environment-Explainable AI (XAI) in autonomous vehicles serves as a bridge between complex technological capabilities and human understanding, addressing three fundamental concerns: safety assurance, regulatory compliance, and public trust.
- Data Oriented XAI
  - Explanations of predictions using ML and Deep Learning

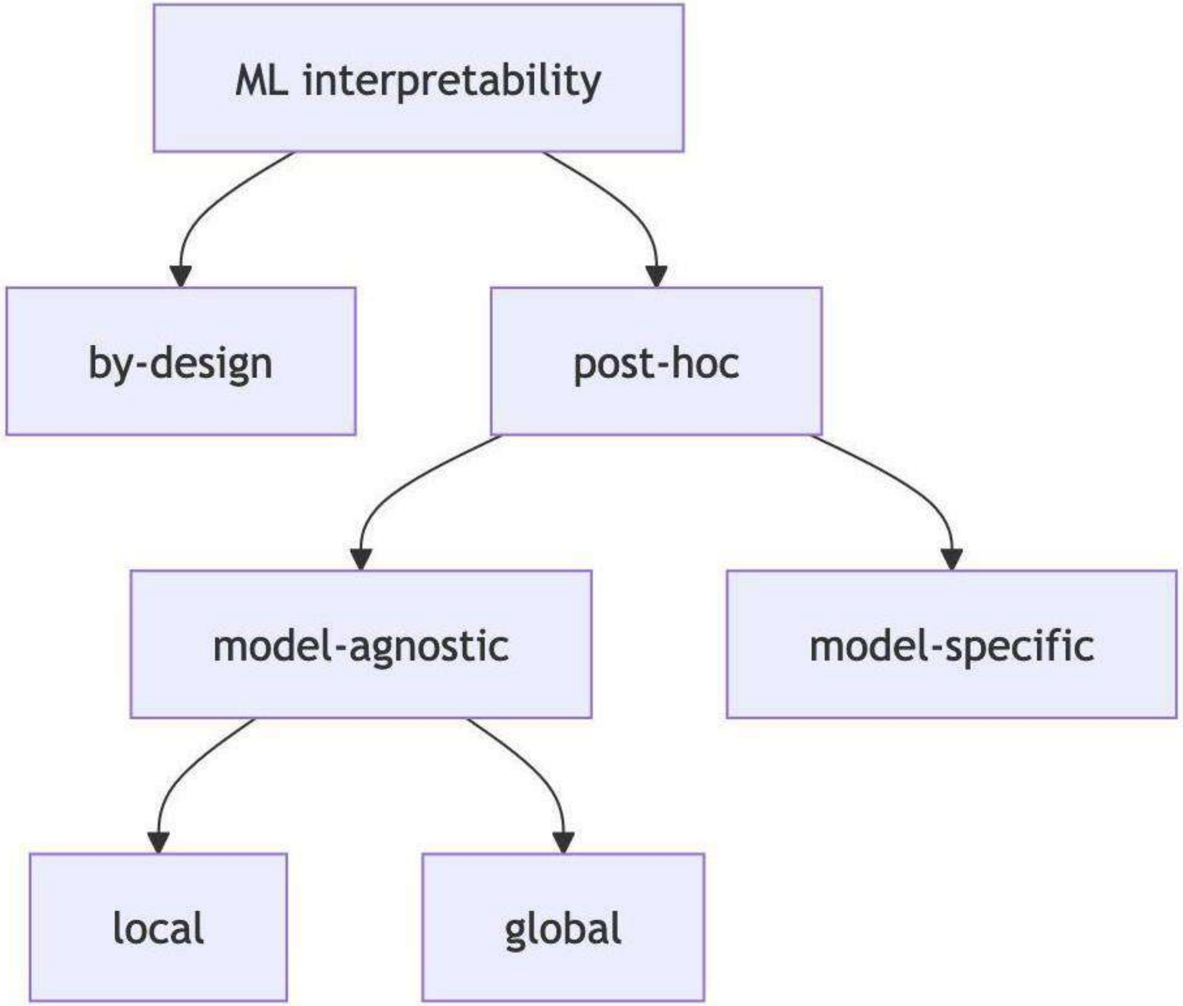
Explainable Models

- Glass Box Models
  - Rule based expert systems
- White Box Models
  - Interpretable ML models
- Black Box Models
  - Deep Learning Models

XAI Taxonomy



Machine Learning Interpretability



Machine Learning Interpretability

- **Interpretability by design** means that we train inherently interpretable models, such as using logistic regression'
- **Post-hoc interpretability** means that we use an interpretability method after the model is trained.
- Post-hoc interpretation methods can be **model-agnostic**, such as permutation feature importance, or **model-specific**, such as analyzing the features learned by a neural network.
- Model-agnostic methods can be further divided into **local** methods which focus on explaining individual predictions, and **global** methods which focus on datasets.

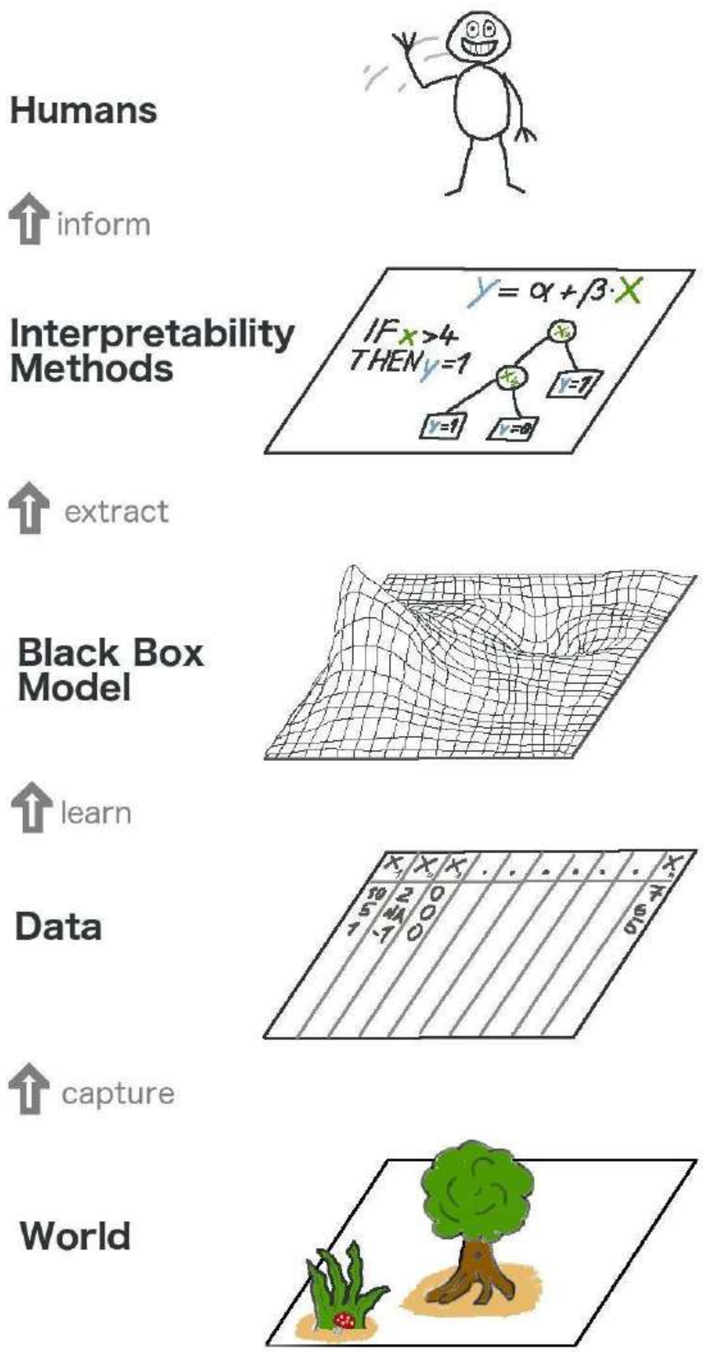
Interpretable models by design

- Models that are interpretable by design are also called intrinsically or inherently interpretable models.

<u>Example</u>

- Linear regression: Fit a linear model by minimizing the sum of squared errors.
- Logistic regression: Extend linear regression for classification using a nonlinear transformation.
- Linear model extensions: Add penalties, interactions, and nonlinear terms for more flexibility.
- Decision trees: Recursively split data to create tree-based models.
- Decision rules: Extract if-then rules from data.
- RuleFit: Combine tree-based rules with Lasso regression to learn sparse rule-based models.

Post-hoc interpretability means that we use an interpretability method after the model is trained.



Model-agnostic methods

- Model-agnostic methods work by the SIPA principle: sample from the data, perform an intervention on the data, get the predictions for the manipulated data, and aggregate the results .
- An example is permutation feature importance.

Causality

- Causality is an influence by which one event, process, state, or object (a cause) contributes to the production of another event, process, state, or object (an effect) where the cause is at least partly responsible for the effect, and the effect is at least partly dependent on the cause.%

Causality

- Causality is the philosophical and scientific concept that describes how one event (the cause) brings about another event (the effect).

Example Questions about Causality:

- Does smoking cause lung cancer?
- Does fertilizer increase crop yield?
- Does an ad increase product sales?

Causality goes beyond correlation (just statistical co-occurrence) to say "X actually produces a change in Y."

Causality Learning (or Causal Learning)

- Causality Learning (or Causal Learning) is the process of automatically discovering or estimating causal relationships from data using algorithms.

- Association: “I noticed X and Y happen together”
- Causuality Learning: “X causes Y — and if I change X, Y will change too”

Causal inference

Causal inference is the process of identifying and understanding cause-effect relationships between variables.

Causality is about interventions, about doing. Whereas standard statistics is about correlations, but they can lead to erroneous assumptions that lead to much worse things. If we start to formalize the use of causality, it is about inferring a treatment or a policy T from an outcome Y.

Causal inference

- Causal inference in machine learning is an important area of research that seeks to improve the ability of machine learning models to capture causal relationships in data.
- This capability may have important implications in a wide variety of fields, and causal inference in machine learning is expected to continue to be an active area of research in the future.

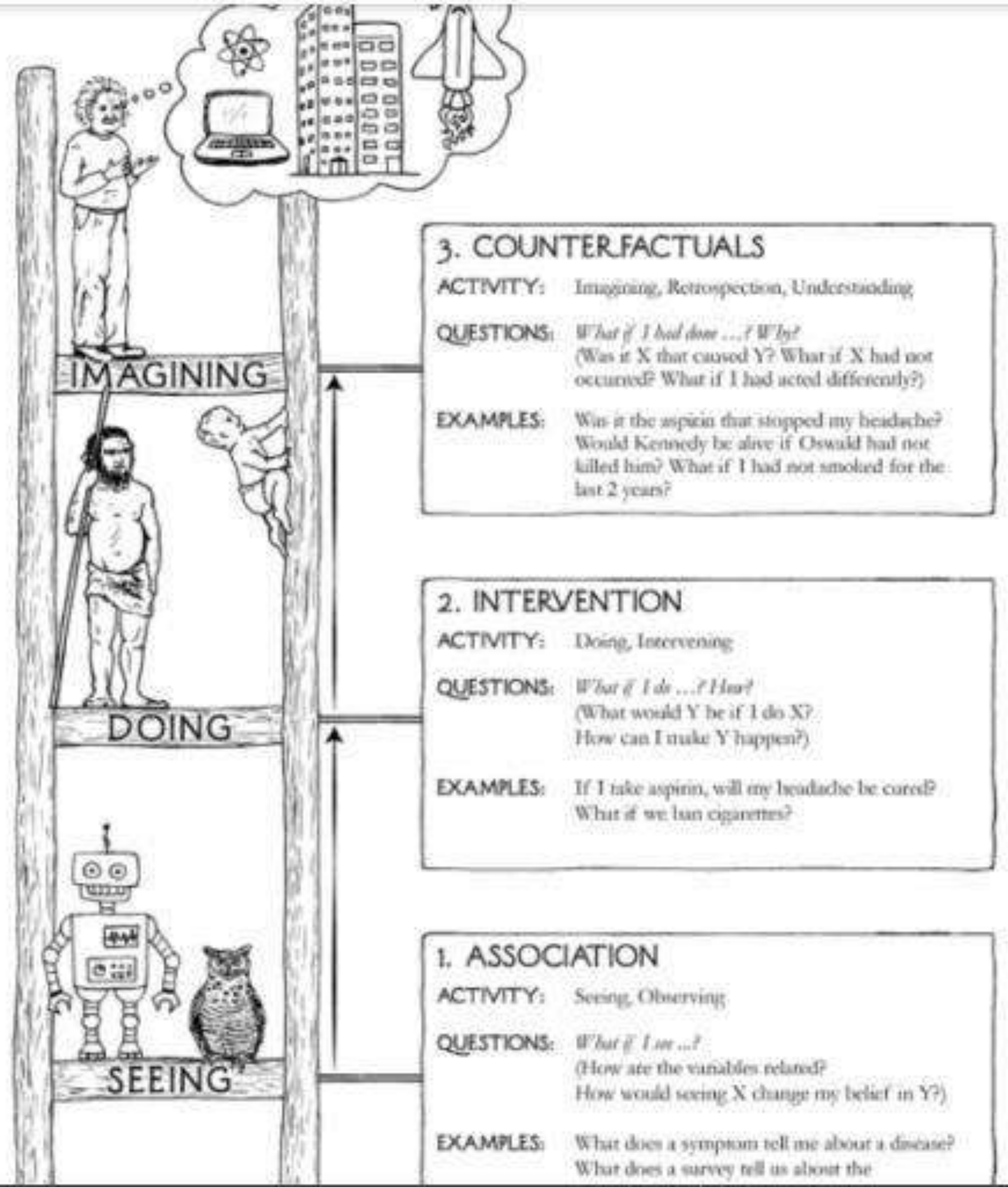
- The goal of causal inference in machine learning is to improve the accuracy and interpretability of models, which can have important implications in areas such as health, economics, policy, and justice.
- For example, causal inference models can be used to understand the effects of interventions and policies, to control for biases in data, and to provide greater transparency and explainability in automated decisions.

Causal inference

- "Would the outcome for this patient have been different if a different medication had been administered?"

The resolution of counterfactual questions presents a significant challenge for two main reasons.

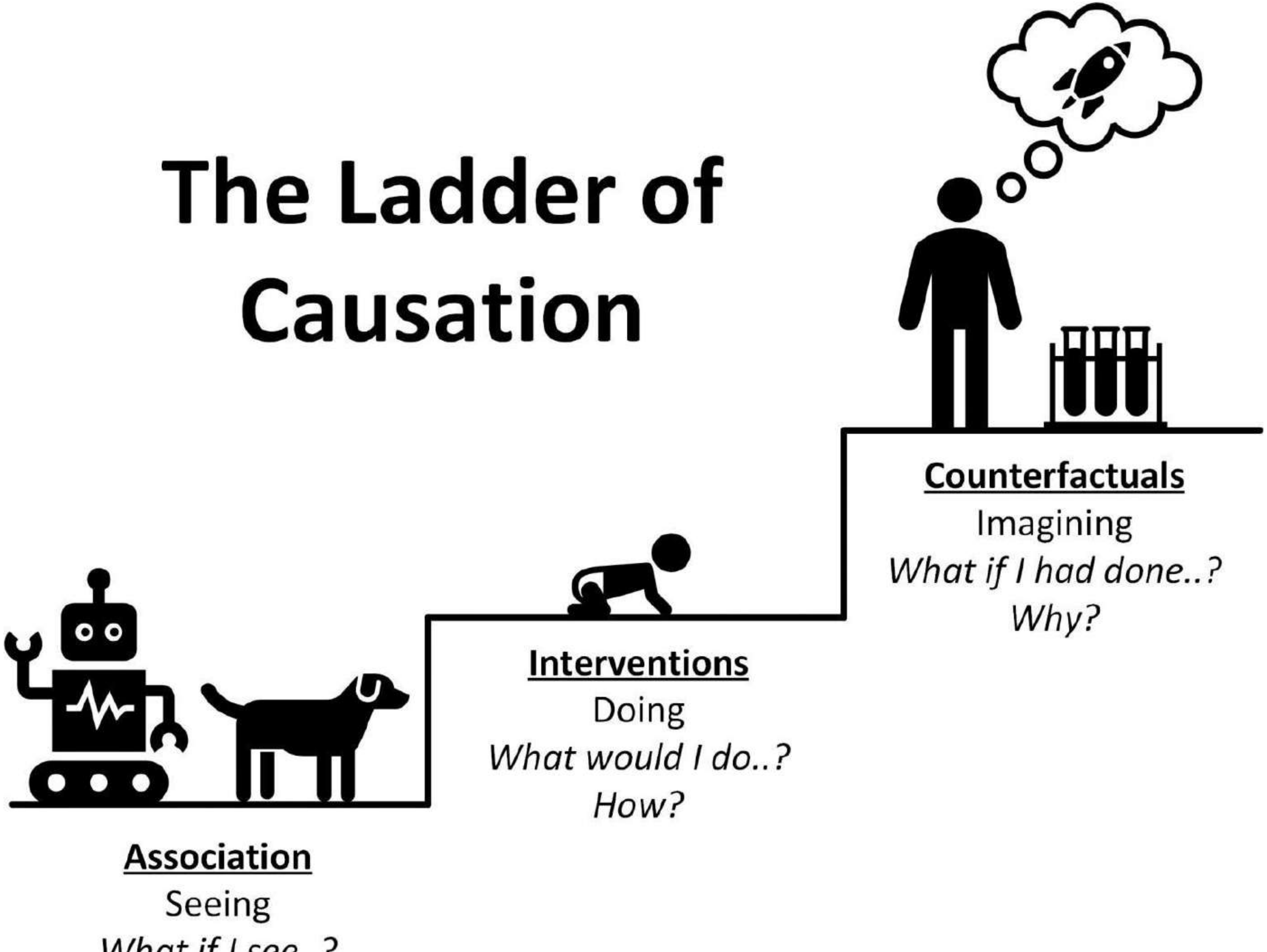
- First, we are only able to observe the actual outcome resulting from the administered treatment, rather than the hypothetical outcomes that could have ensued under alternate treatments.
- Second, in observational studies, treatment assignments are often non-random, which creates differences between the treatment group and the population. This key challenge arises from the inability to observe both counterfactual outcomes and actual outcomes for an individual under different treatment scenarios at the same time.



The Book of Why

The New Science of Cause and Effect

By [Judea Pearl](), [Dana Mackenzie]() ·


The Ladder of Causation
Counterfactuals
Imagining
What if I had done..?
Why?
Interventions
Doing
What would I do..?
How?
Association
Seeing

Ladder of causation

1. Association

a. Action: The child notices that after the plant is watered, the withering leaves appear revitalized.

b. Causal Reasoning: The child associates watering with improving the plant's appearance.

Ladder of causation

2. Intervention

a. Action: The child decides to water the plant themselves once they see the leaves begin to wither.

b. Causal Reasoning: The child learns a cause-and-effect relationship. The intervention (watering) consistently results in an effect (plant appearance).

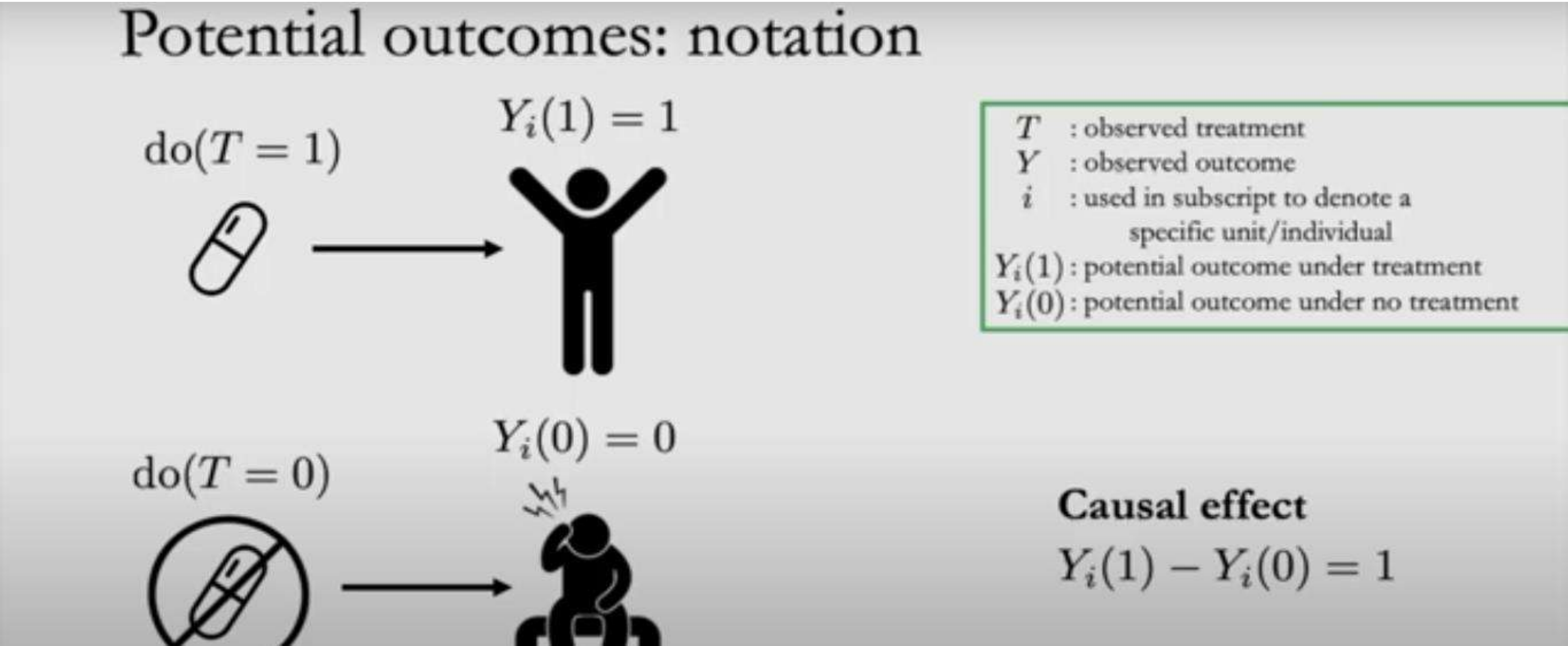
Ladder of causation

3. Counterfactual

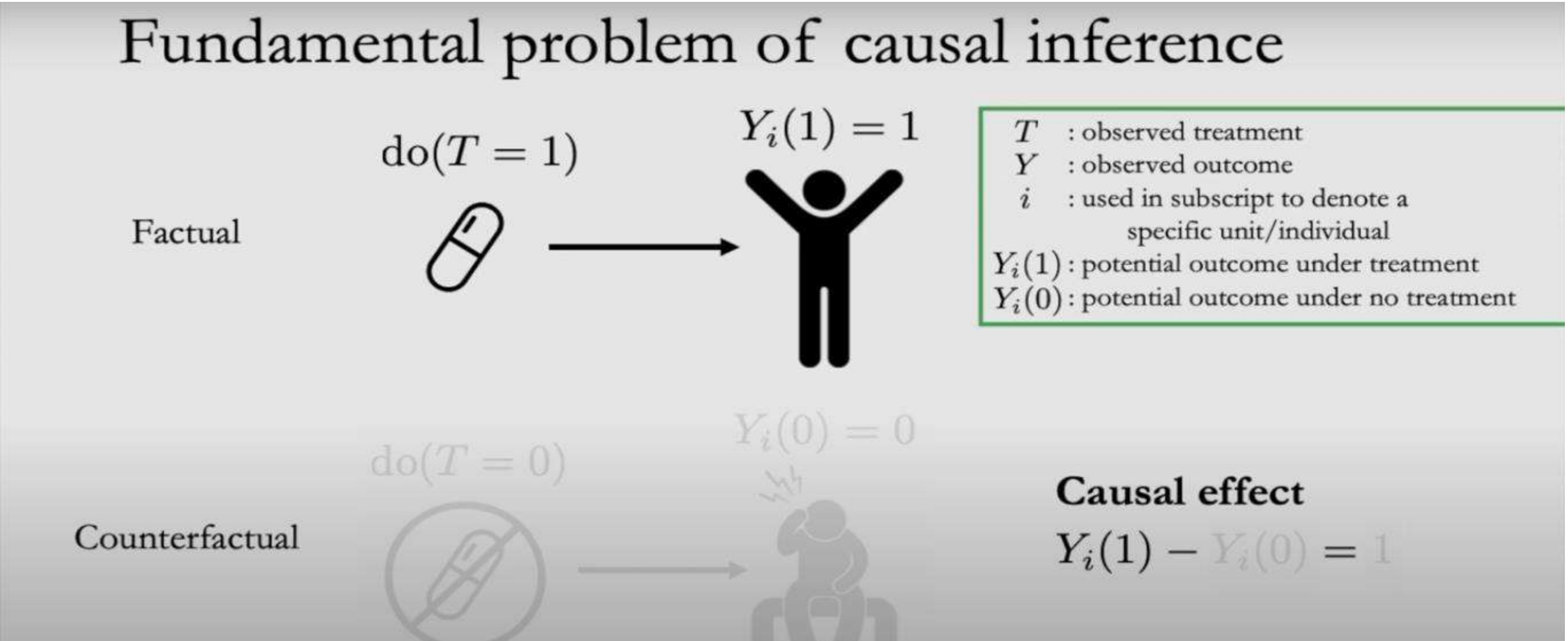
a. Action: The child wonders, “What would happen to the plant if it was never watered?”

b. Causal Reasoning: Without direct testing, the child can infer that if they do not water the plant, its health would continue to decline, and it would die. This counterfactual thinking reinforces the child’s understanding of the importance of watering for the plant’s well-being.

Counterfactual



Counterfactual



A unit is the basic element of analysis in studies of treatment effects. It can be an individual, an entity, or any observational unit in a dataset or a study. Units can be a range of entities that are observed, such as people, animals, households, regions, and others, depending on the research question.

Treatment is the variable, action, or intervention whose causal effect on an outcome is the focus of the study.

Units are categorized into two groups: the treated group and the control group. The treated group includes units subjected to the treatment or intervention under investigation, while the control group comprises units that do not receive treatment.

The potential outcome refers to the outcome of the units that receive the treatment, essentially, the outcome of the treated group.

The observed or factual outcome is the outcome that occurs when a specific treatment, action, or intervention is applied. It is the real outcome observed in a group of units following a specific treatment or intervention.

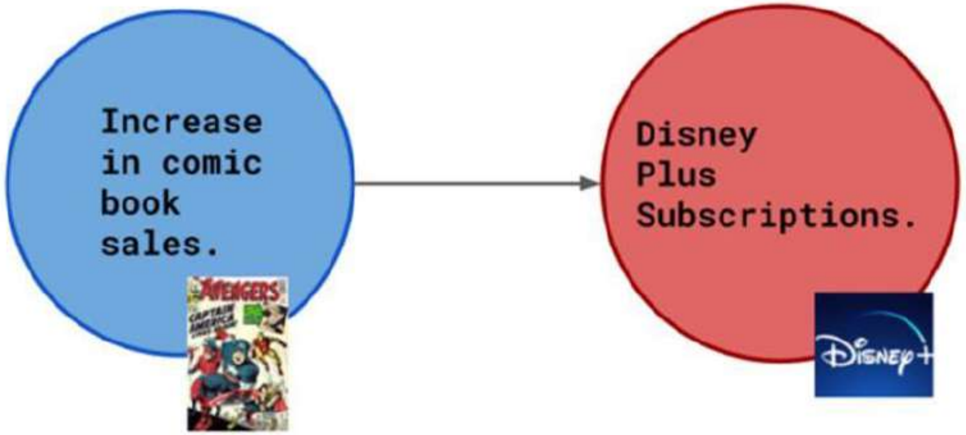
The counterfactual outcome is the result of a unit having received a different treatment. It imagines the outcome for the same group of units had they not received the actual treatment or intervention under study.

Example

- There has been a recent rise in comic book sales. After some data analysis, you conclude that there is a direct correlation between comic book sales and Disney+ subscriptions. But you still don't have an exact reason, so you develop some scenarios to form a hypothesis.

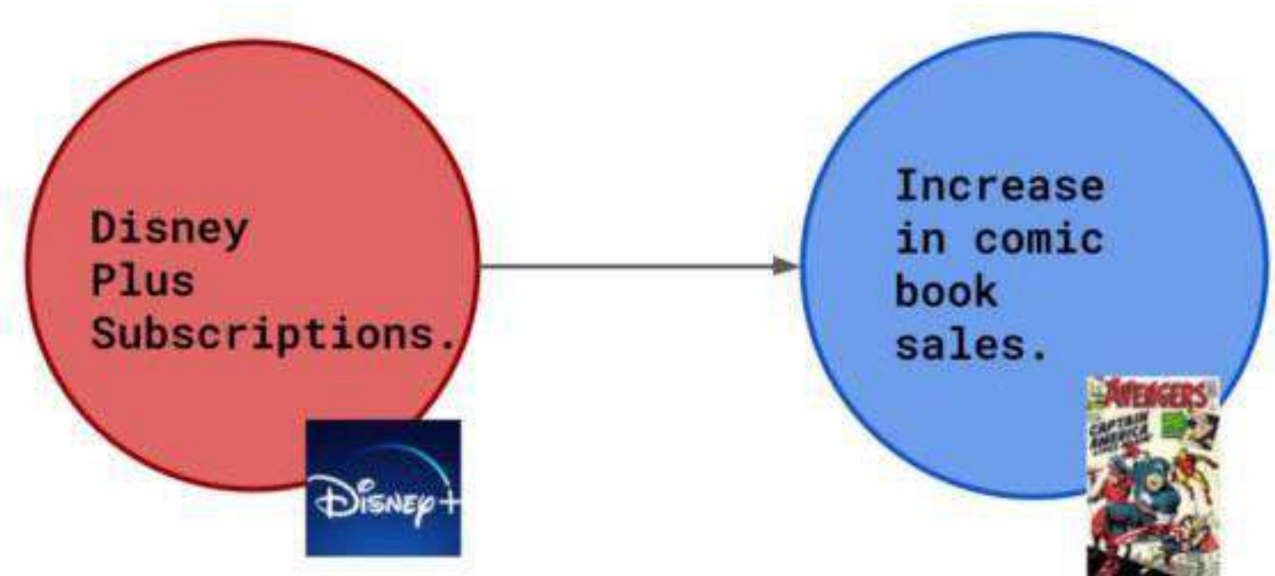
Scenario 1-Direct Cause

- Scenario 1, where increased comic book sales cause more Disney+ subscriptions ,
- The comics display ads from Disney+, so increased comic book sales cause more Disney+ subscriptions.



Scenario 2-Reversing Cause and Effect

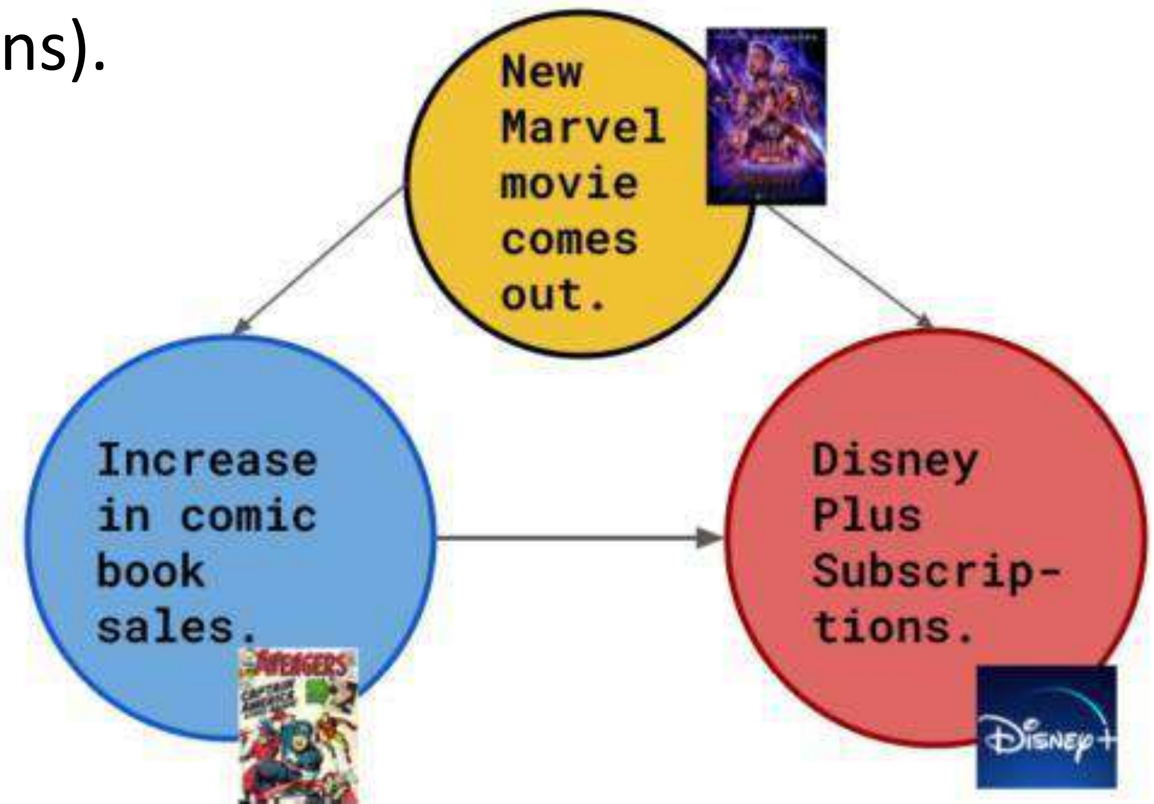
- Scenario 2, where an increase in Disney+ subscriptions causes an increase in comic book sales. Disney+ has shows about comic book characters. New fans would like to consume more of such content. So an increase in Disney+ subscriptions led to more Marvel Comics being bought.



Scenario 3-Investigating a Hidden cause

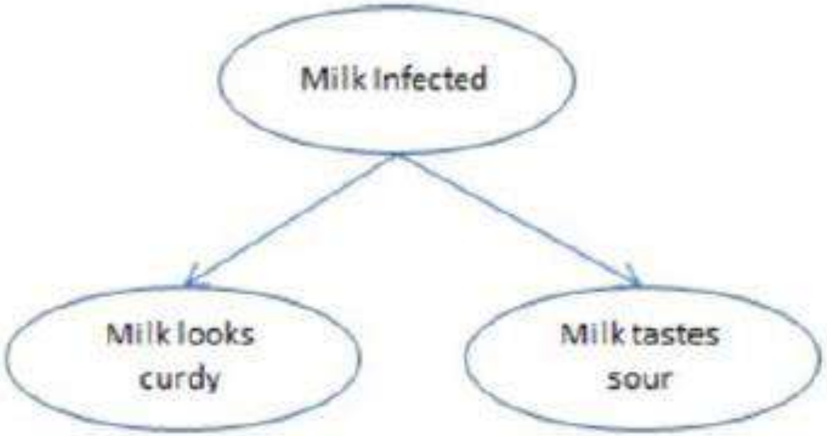
Scenario 3, where a hidden cause affects the outcome (source: image by the author). Each node is a variable, and each arrow shows the direction of the causal connection.

All Marvel Cinematic Universe movies are on Disney+. When a new Marvel movie comes out, the hype around these characters leads to more comic book sales (and, in turn, more Disney+ subscriptions).

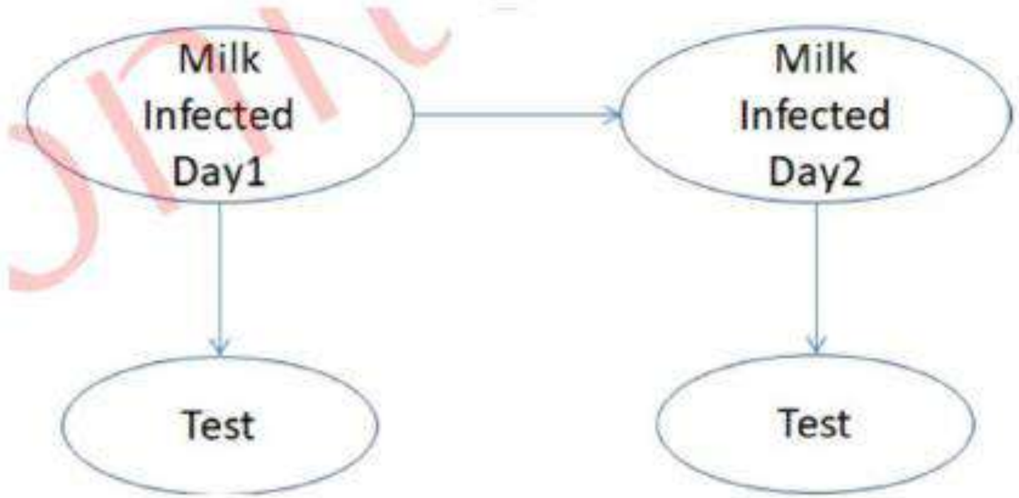


Causal Model Induction (Causality learning)

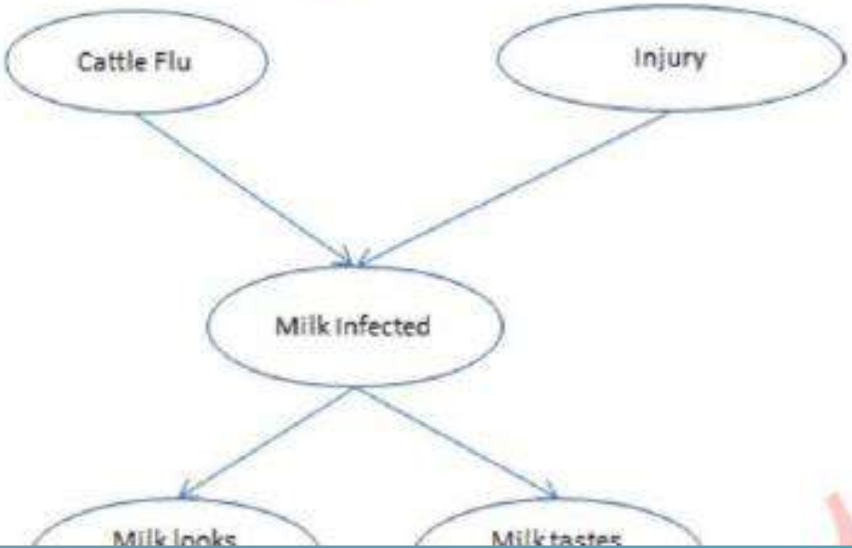
- Integrates causal reasoning into the explanation of AI models
- Helps to identify **cause-and-effect relationships** from data, rather than just correlations.
- This is crucial for understanding how interventions influence outcomes.
- **Causal Effect Estimation**
  - Cause estimation with hidden confounders
  - Cause estimation with interference
  - Cause estimation with graph structured treatments
- Causal models and decision trees – foundation of XAI ---> Recommendations
  - **Ex1: Estimation of New drug effect on different dosage**
  - **Ex2: Estimation of cause of infected milk**



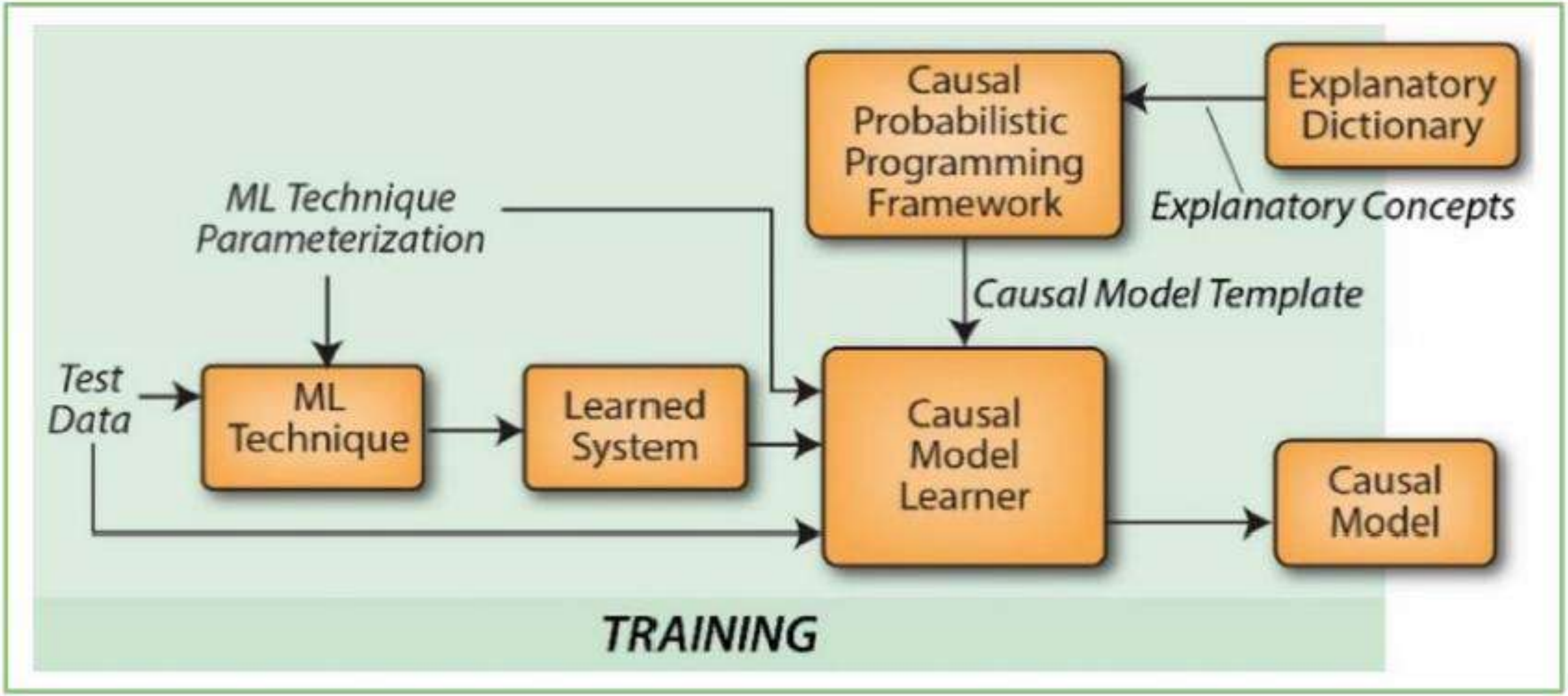
- Milk Infection being the confounder for the test1



6.6: Interference of the condition of the milk on the previous day



Causal Model Induction



Experiment with the learned model to learn an explainable, causal,

Causal Machine Learning

- Causal ML, an emerging area of research that seeks to improve the ability of machine learning models to capture causal relationships in data.

- Causal inference in machine learning is based on the idea that correlations between variables are often not sufficient to establish causal relationships, as there may be other variables that influence both.

- Machine learning models often rely on correlation learning, i.e., the ability to find patterns in data to make predictions. However, this capability can be limited in situations where a deeper understanding of the underlying causal relationships is required.

- Causal inference in machine learning seeks to address this limitation by using techniques and algorithms that take into account the causal relationships between variables.

Causal models

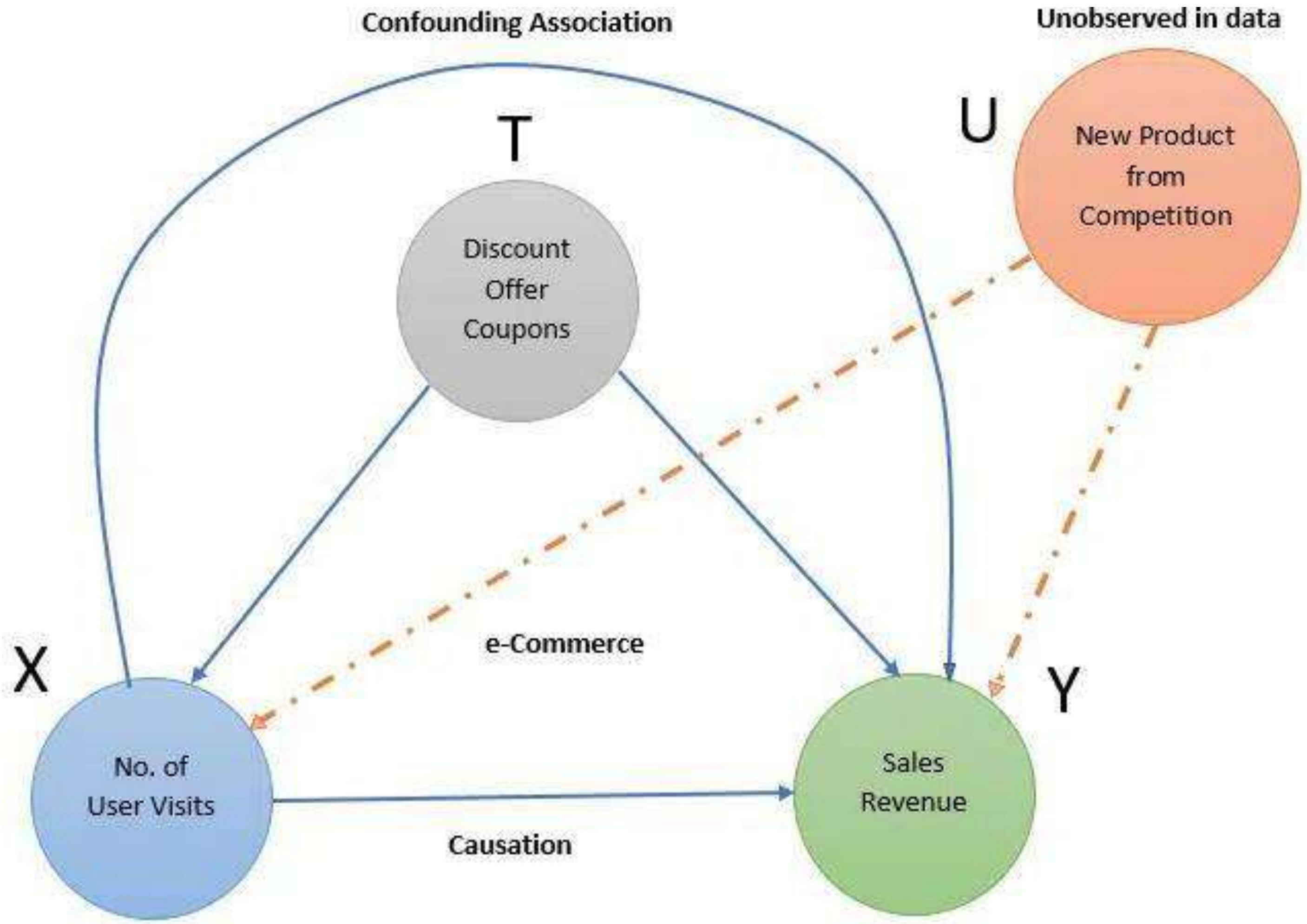
- **Prediction models are bidirectional in nature** (e.g.: Given X, we can predict Y, or Given Y, we can predict X), whereas causal models have a direction and relationship between X and Y (X causes Y and is unidirectional).
- Prediction focuses on What, whereas causal focuses on Why. They can complement each other.
- Causal Learning involves creating models that capture the causal mechanism based on the observed data, a new set of variables, and assumptions and incorporate the changes in data distribution when subject to different interventions.

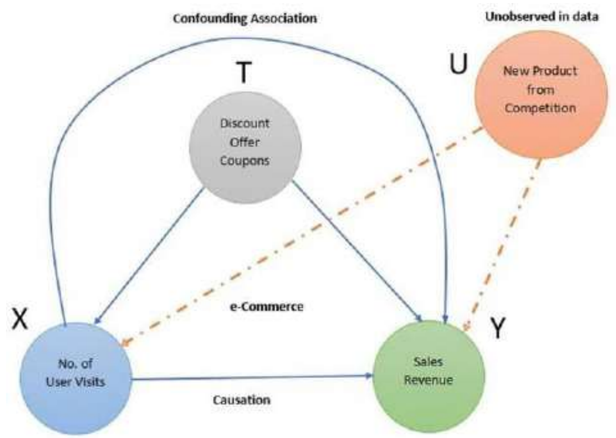
Examples

Causal learning can be applied in determing —

- Effect of various treatments for a disease
- Effect of marketing campaigns on sales
- Effect of salary increments on employee retention
- Effect of customer satisfaction on improved service
- Effect of new product features on customer adoption

Representation of a causal relationship in an e-commerce scenario.





- In this image, we are looking at the effect of intervention X on Y.
- While in traditional machine learning, we directly look at the correlation between the variables and outcomes, in causality we look for the effect of intervention T on the variable X and how it impacts the outcome Y.
- Here intervention is a confounding variable, T.
- Failing to account for confounding variables can lead to wrong conclusions on the relationship between independent and dependent variables.
- Another interesting factor here is the influence of a new product launch from the competition, U. In traditional machine learning, we treat many of these extraneous factors as noise or errors that are then ignored.
- With causal inference we would be able to compute the ATE (Average Treatment Effect) or CATE (Conditional Average Treatment Effect) and measure the impact of an intervention on the outcome – For e.g., an Increase in Sales revenue due to a discount offer vs Sales revenue with no offer.

XAI Limitations

- Trade-off Between Fidelity and Simplicity
- Subjectivity in Understandability
- Incomplete Explanations
- Information Overload
- Computational Cost
- Security and Privacy Risks

References

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